

Vishal Gopinath

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AI Engineer with 4+ years of experience building Generative AI, Deep Learning, and NLP systems in Python across finance and customer operations. Architected RAG knowledge platforms using OpenAI APIs and open-source LLMs with LangChain, LangGraph, and Weaviate, reducing hallucinated responses by 50%+. Designed multilingual Voice AI agents that automated ~70% of CSR calls across 500+ daily customer interactions. Deployed GPU-backed LLM and Machine Learning services on Kubernetes with Docker and LangSmith observability at enterprise scale.

WORK EXPERIENCE

MSI Services Private Limited

Bengaluru, India

AI Developer II

04/2025 - Present

- Led enterprise-wide LLM initiatives across finance, operations, sales, and customer service, reducing manual effort by **40–60%** across key workflows.
- Architected **RAG-based knowledge systems using OpenAI APIs and Weaviate**, improving enterprise query accuracy and reducing hallucinated responses by **50%+**.
- Built internal and customer-facing AI chatbots for product queries, order placement, dealer lookup, stock availability, ETA tracking, and delivery updates, improving response turnaround by **70%+**.
- Designed multilingual **Voice AI agents** for inbound support and outbound workflows including order confirmation, delivery reminders, payment follow-ups, and customer/store communication, automating **~70% of CSR calls**.
- Developed an AI-driven **call quality evaluation pipeline** using STT and conversational analytics to assess agent conversations across **English, French, and Spanish**, eliminating full manual QA for **10+ CSR team members**.
- Deployed **GPU-backed LLM services on Kubernetes** with LangSmith observability, improving production monitoring, reliability, and cost optimization.

AI Developer

04/2023 - 03/2025

- Automated **invoice** and **check deposit workflows** using **OCR + NLP** extraction, improving efficiency by **~85%** and reducing reconciliation errors.
- Built external AI **conversational systems (LangChain, LangGraph)** handling **500+** daily customer queries with intelligent escalation.
- Developed real-time **RAG-powered AI assistant** supporting CSRs during live calls.
- Architected self-learning **product search & ranking pipeline** processing 10,000+ orders/day, leveraging behavioral signals and feedback loops to continuously improve product selection accuracy.
- Integrated AI services with enterprise backend systems via **REST APIs**, eliminating manual workflows.

- Developed **DSP** and **speech enhancement pipelines** using **NumPy & SciPy**, improving noise reduction accuracy (**63% → 81%**).
- Implemented **MFCC & PPG feature extraction**; trained **ML** models achieving **~90%** classification accuracy.

EDUCATION

M. Tech. in Data Science & AI

PES University • GPA: 9.8 Bengaluru • 07/2024 – 08/2026

B. Tech. in Electronics and Communication Engineering

JSS Academy of Technical Education • GPA: 9.17 Bengaluru • 03/2018 – 06/2022

SKILLS

Core Expertise: Generative AI, RAG Systems, Conversational AI, Search & Ranking Models, NLP, Deep Learning, Multi-Agent Systems

LLMs & APIs: OpenAI APIs, Claude, Gemini, Open-source LLMs (LLaMA | Mistral | Falcon | Phi-3 | Gemma), Whisper

Data & Infrastructure: Python, Weaviate, Pinecone, Elasticsearch, MySQL, SQL, Docker, Kubernetes, REST APIs, LangChain, LangGraph, LangSmith, Weights & Biases

CERTIFICATIONS

Career Essentials in Generative AI by Microsoft and LinkedIn 07/2024
Microsoft

Advance your skills in Natural Language Processing 08/2023
LinkedIn

Advanced Python & DSA 03/2023
LinkedIn

Prepare for Google Cloud Platform Associate Cloud Engineer 03/2023
LinkedIn

Advance your skills in Deep Learning and Neural Networks 02/2023
LinkedIn

PROJECTS

Meta-Cognitive Neuro-Symbolic Learning 04/2026 – 09/2026
M.Tech Academic Project

- Developed an AI trust evaluation framework to detect when models produce correct predictions using biased patterns or shortcut reasoning.
- Restored and executed a research pipeline to assess model accuracy, concept understanding, confidence reliability, and performance under data shifts.

- Compared multiple confidence estimation methods to identify overconfident or unreliable AI behavior.
- Analyzed experimental results showing that trustworthy AI requires both accurate outputs and reliable internal reasoning.
- Proposed a meta-cognitive confidence layer to flag predictions that may appear correct but are not logically trustworthy.

AI Model to Analyze Sleep Quality

10/2025 - 03/2026

M.Tech Academic Project

- Designed and implemented an **end-to-end machine learning pipeline** to **predict sleep quality** using structured health and lifestyle data.
- Performed data preprocessing, feature engineering, model selection, and hyperparameter tuning across multiple classification algorithms.
- Achieved **88%** prediction accuracy through systematic experimentation and performance optimization.
- Leveraged **Model Cloud GPU instances** for accelerated training and scalable experimentation.
- Integrated **Weights & Biases (W&B)** for experiment tracking, metric logging, and model performance monitoring across training runs.